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Signal processing device, signal processing method, and distance-measuring module

Abstract

There is provided a signal processing device, a signal processing method, and a distance-measuring module that allow improvement in distance measurement accuracy. The signal processing device includes an estimation unit that estimates, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light. The present technology can be applied to, for example, a distance-measuring module that performs distance measurement by the indirect ToF method, and the like.

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Background/Summary

CROSS REFERENCE TO PRIOR APPLICATION
(1) This application is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2020/012199 (filed on Mar. 19, 2020) under 35 U.S.C. § 371, which claims priority to Japanese Patent Application No. 2019-072688 (filed on Apr. 5, 2019), which are all hereby incorporated by reference in their entirety.
TECHNICAL FIELD
(2) The present technology relates to a signal processing device, a signal processing method, and a distance-measuring module, and in particular to a signal processing device, a signal processing method, and a distance-measuring module that allow improvement in distance-measuring accuracy.
BACKGROUND ART
(3) In recent years, because of an advance in the semiconductor technology, distance-measuring modules that measure the distance to an object have become increasingly small. Therefore, for example, it has been implemented to mount a distance-measuring module in a mobile terminal such as a so-called smartphone, which is a small information processing device equipped with a communication function.
(4) Examples of a distance measuring method in the distance-measuring module include the indirect time of flight (ToF) method, the structured light method, and the like. The indirect ToF method includes emitting light toward an object, detecting light reflected from a surface of the object, and calculating the distance to the object on the basis of a measured value obtained by measuring the time of flight of the light. The structured light method includes emitting pattern light toward an object and calculating the distance to the object on the basis of an image obtained by capturing distortion of the pattern on the surface of the object.
(5) For example, Patent Document 1 discloses a technology for accurately measuring a distance by determining movement of an object within a detection period with the distance-measuring module for distance measurement by the indirect ToF method.
CITATION LIST
Patent Document
(6) Patent Document 1: Japanese Patent Application Laid-Open No. 2017-150893
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
(7) In the distance-measuring module of the indirect ToF method, further improvement in distance

measurement accuracy is required.
(8) The present disclosure has been made in view of such a situation, and makes it possible to improve the distance measurement accuracy.
Solutions to Problems
(9) A signal processing device according to a first aspect of the present technology includes an estimation unit that estimates, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(10) A signal processing method according to a second aspect of the present technology includes, by a signal processing device, estimating, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(11) A distance-measuring module according to a third aspect of the present technology includes: a light-receiving unit in which pixels each including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit are two-dimensionally arranged; and a signal processing unit including an estimation unit that estimates a sensitivity difference between taps of the first tap and the second tap in the pixels by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(12) According to the first to third aspects of the present technology, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap is estimated by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(13) The signal processing device and the distance-measuring module may be independent devices or modules incorporated in another device.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a block diagram showing a configuration example of one embodiment of a distance-measuring module to which the present technology is applied.
(2) FIG. **2** is a diagram for describing an operation of a pixel in an indirect ToF method.
(3) FIG. **3** is a diagram for describing a detection method by 4 phase.
(4) FIG. **4** is a diagram for describing the detection method by 4 phase.
(5) FIG. **5** is a diagram for describing a method of calculating a depth value by the 2-phase method and the 4-phase method.
(6) FIG. **6** is a diagram for describing driving of a light-receiving unit of the distance-measuring module and output timing of a depth map.
(7) FIG. **7** is a block diagram showing a detailed configuration of a signal processing unit.
(8) FIG. **8** is a diagram for describing detection signals of four phases.

(9) FIG. **9** is a diagram for describing a blend rate based on a movement amount and amplitude.
(10) FIG. **10** is a block diagram showing a detailed configuration example of a fixed pattern estimation unit.
(11) FIG. **11** is a diagram for describing the blend rate of coefficient updates based on the movement amount and amplitude.
(12) FIG. **12** is a flowchart for describing depth value calculation processing on a pixel to be processed by the signal processing unit.
(13) FIG. **13** is a diagram for describing a first modification and a second modification of driving.
(14) FIG. **14** is a diagram for describing the first modification and the second modification of driving.
(15) FIG. **15** is a diagram for describing a third modification of driving.
(16) FIG. **16** is a block diagram showing a configuration example of an electronic device to which the present technology is applied.
(17) FIG. **17** is a block diagram showing a configuration example of one embodiment of a computer to which the present technology is applied.
(18) FIG. **18** is a block diagram showing one example of a schematic configuration of a vehicle control system.
(19) FIG. **19** is an explanatory diagram showing one example of installation positions of an outside vehicle information detection unit and an image capturing unit.

MODE FOR CARRYING OUT THE INVENTION

(20) A mode for carrying out the present technology (hereinafter referred to as an embodiment) will be described below. Note that the description will be made in the following order. 1. Configuration example of distance-measuring module 2. Pixel operation of indirect ToF method 3. Output timing of depth map 4. Detailed configuration example of signal processing unit 5. Depth value calculation processing of signal processing unit 6. Modifications of driving by distance-measuring module 7. Configuration example of electronic device 8. Configuration example of computer 9. Example of application to mobile object

<1. Configuration Example of Distance-Measuring Module>

(21) FIG. **1** is a block diagram showing a configuration example of one embodiment of a distance-measuring module to which the present technology is applied.
(22) The distance-measuring module **11** shown in FIG. **1** is a distance-measuring module that measures a distance by the indirect ToF method, and includes a light-emitting unit **12**, a light emission control unit **13**, a light-receiving unit **14**, and a signal processing unit **15**. The distance-measuring module **11** emits light to an object and receives light (reflected light) obtained by the light (emission light) reflected by the object, thereby measuring a depth value as distance information to the object and outputting a depth map.
(23) The light-emitting unit **12** includes, for example, an infrared laser diode and the like as a light source, emits light while performing modulation at timing according to a light emission control signal supplied from the light emission control unit **13** according to control by the light emission control unit **13**, and emits emission light to the object.
(24) The light emission control unit **13** supplies the light emission control signal having a predetermined frequency (for example, 20 MHz and the like) to the light-emitting unit **12**, thereby controlling the light emission of the light-emitting unit **12**. Furthermore, in order to drive the light-receiving unit **14** in accordance with the timing of light emission in the light-emitting unit **12**, the light emission control unit **13** also supplies the light emission control signal to the light-receiving unit **14**.
(25) In the light-receiving unit **14**, a pixel array unit **22** is provided in which pixels **21** that each generate a charge according to a received light amount and output a signal according to the charge are two-dimensionally arranged in a matrix in the row direction and the column direction, and a drive control circuit **23** is disposed in a peripheral region of the pixel array unit **22**.

(26) The light-receiving unit **14** receives the reflected light from the object by using the pixel array unit **22** in which the plurality of pixels **21** is two-dimensionally arranged. Then, the light-receiving unit **14** supplies the signal processing unit **15** with pixel data including detection signals according to the received light amount of the reflected light received by each pixel **21** of the pixel array unit **22**.
(27) The signal processing unit **15** calculates the depth value, which is the distance from the distance-measuring module **11** to the object, for each pixel **21** of the pixel array unit **22** on the basis of the pixel data supplied from the light-receiving unit **14**, and outputs the depth value to a subsequent control unit (for example, application processing unit **121**, operation system processing unit **122**, and the like of FIG. **16**). Alternatively, the signal processing unit **15** may generate the depth map in which the depth value is stored as a pixel value of each pixel **21** of the pixel array unit **22**, and output the depth map to the subsequent stage. Note that the detailed configuration of the signal processing unit **15** will be described later with reference to FIG. **7**.
(28) The drive control circuit **23** outputs a control signal for controlling driving of the pixel **21** (for example, distribution signal DIMIX, selection signal ADDRESS DECODE, reset signal RST, and the like to be described later), for example, on the basis of the light emission control signal supplied from the light emission control unit **13**, and the like.
(29) The pixel **21** includes a photodiode **31**, and a first tap **32**A and a second tap **32**B that detect the charge photoelectrically converted by the photodiode **31**. In the pixel **21**, the charge generated by one photodiode **31** is distributed to the first tap **32**A or the second tap **32**B. Then, out of the charge generated by the photodiode **31**, the charge distributed to the first tap **32**A is output from a signal line **33**A as a detection signal A, and the charge distributed to the second tap **32**B is output from a signal line **33**B as a detection signal B.
(30) The first tap **32**A includes a transfer transistor **41**A, a floating diffusion (FD) unit **42**A, a selection transistor **43**A, and a reset transistor **44**A. Similarly, the second tap **32**B includes a transfer transistor **41**B, an FD unit **42**B, a selection transistor **43**B, and a reset transistor **44**B.
(31) <2. Pixel Operation of Indirect ToF Method>
(32) The operation of the pixel **21** by the indirect ToF method will be described with reference to FIG. **2**.
(33) As shown in FIG. **2**, the emission light modulated to repeat emission on/off at emission time T (1 cycle=2T) is output from the light-emitting unit **12**, and the reflected light is received by the photodiode **31** with delay time ΔT according to the distance to the object. Furthermore, the distribution signal DIMIX_A controls on/off of the transfer transistor **41**A, and the distribution signal DIMIX_B controls on/off of the transfer transistor **41**B. The distribution signal DIMIX_A is a signal having the same phase as the emission light, and the distribution signal DIMIX_B has a phase obtained by inverting the distribution signal DIMIX_A.
(34) Therefore, the charge generated when the photodiode **31** receives the reflected light is transferred to the FD unit **42**A while the transfer transistor **41**A is on in response to the distribution signal DIMIX_A, and is transferred to the FD unit **42**B while the transfer transistor **41**B is on in response to the distribution signal DIMIX_B. With this configuration, in a predetermined period in which the emission of the emission light of the emission time T is periodically performed, the charge transferred via the transfer transistor **41**A is sequentially accumulated in the FD unit **42**A, and the charge transferred via the transfer transistor **41**B is sequentially accumulated in the FD unit **42**B.
(35) Then, after the period for accumulating the charge is finished, when the selection transistor **43**A is turned on in response to the selection signal ADDRESS DECODE_A, the charge accumulated in the FD unit **42**A is read via the signal line **33**A, and the detection signal A according to the charge amount is output from the light-receiving unit **14**. Similarly, when the selection transistor **43**B is turned on in response to the selection signal ADDRESS DECODE_B, the charge accumulated in the FD unit **42**B is read via the signal line **33**B, and the detection signal

B according to the charge amount is output from the light-receiving unit **14**. Furthermore, the charge accumulated in the FD unit **42**A is discharged when the reset transistor **44**A is turned on in response to the reset signal RST_A, and the charge accumulated in the FD unit **42**B is discharged when the reset transistor **44**B is turned on in response to the reset signal RST_B.

(36) In this way, the pixel **21** distributes the charge generated by the reflected light received by the photodiode **31** to the first tap **32**A or the second tap **32**B according to the delay time ΔT, and outputs the detection signal A and the detection signal B. Then, the delay time ΔT corresponds to the time during which the light emitted by the light-emitting unit **12** flies to the object, is reflected by the object, and then flies to the light-receiving unit **14**, that is, corresponds to the distance to the object. Therefore, the distance-measuring module **11** can obtain the distance to the object (depth value) according to the delay time ΔT on the basis of the detection signal A and the detection signal B.

(37) However, in the pixel array unit **22**, because of discrepancy in characteristics (sensitivity difference) of each element of the pixel transistor of the individual pixel **21** including the photodiode **31**, the transfer transistor **41**, and the like, influences different from pixel **21** to pixel **21** may be exerted on the detection signal A and the detection signal B. Therefore, the distance-measuring module **11** of the indirect ToF method adopts a method of removing the sensitivity difference between taps as fixed pattern noise of each pixel and improving the SN ratio by acquiring the detection signal A and the detection signal B obtained by receiving the reflected light by the same pixel **21** with phases changed.

(38) As a method of receiving reflected light by changing the phase and calculating the depth value, for example, the detection method by 2 phase (2-phase method) and the detection method by 4 phase (4-phase method) will be described.

(39) As shown in FIG. **3**, the light-receiving unit **14** receives the reflected light at reception timing with the phase shifted by 0°, 90°, 180°, and 270° with respect to the emission timing of the emission light. More specifically, the light-receiving unit **14** receives the reflected light by changing the phase in a time division manner, such as receiving light with the phase set at 0° with respect to the emission timing of the emission light in one frame period, receiving light with the phase set at 90° in the next frame period, receiving light with the phase set at 180° in the next frame period, and receiving light with the phase set at 270° in the next frame period.

(40) FIG. **4** is a diagram showing the light receiving period (exposure period) of the first tap **32**A of the pixel **21**, one above the other, in each phase of 0°, 90°, 180°, and 270° to make the phase difference easy to understand.

(41) As shown in FIG. **4**, in the first tap **32**A, the detection signal A obtained by receiving light in the same phase as the emission light (phase 0°) is called detection signal A**0**, the detection signal A obtained by receiving light in the phase shifted by 90 degrees from the emission light (phase 90°) is called detection signal A**1**, the detection signal A obtained by receiving light in the phase shifted by 180 degrees from the emission light (phase 180°) is called detection signal A**2**, and the detection signal A obtained by receiving light in the phase shifted by 270 degrees from the emission light (phase 270°) is called detection signal A**3**.

(42) Furthermore, although illustration is omitted, in the second tap **32**B, the detection signal B obtained by receiving light in the same phase as the emission light (phase 0°) is called detection signal B**0**, the detection signal B obtained by receiving light in the phase shifted by 90 degrees from the emission light (phase 90°) is called detection signal B**1**, the detection signal B obtained by receiving light in the phase shifted by 180 degrees from the emission light (phase 180°) is called detection signal B**2**, and the detection signal B obtained by receiving light in the phase shifted by 270 degrees from the emission light (phase 270°) is called detection signal B**3**.

(43) FIG. **5** is a diagram for describing a method of calculating a depth value d by the 2-phase method and the 4-phase method.

(44) In the indirect ToF method, the depth value d can be obtained by the following Formula (1).

[Formula 1]

(45) $$d = \frac{c \text{.Math.} \quad T}{2} = \frac{c \text{.Math.}}{4 \quad f} \quad (1)$$

(46) In Formula (1), c is the speed of light, ΔT is the delay time, and f is the modulation frequency of light. Furthermore, φ in Formula (1) represents the phase shift amount [rad] of the reflected light, and is represented by the following Formula (2).

[Formula 2]

(47) $$= \arctan\left(\frac{Q}{I}\right)(0 \leq \quad > 2 \quad) \quad (2)$$

(48) By the 4-phase method, I and Q of Formula (2) are calculated by the following Formula (3) by using the detection signals A**0** to A**3** and the detection signals B**0** to B**3** obtained by setting the phase at 0°, 90°, 180°, and 270°. I and Q are signals obtained by assuming that the luminance change in the emission light is a cos wave and converting the phase of the cos wave from polar coordinates to the rectangular coordinate system (IQ plane).

$I = c_{0} - c_{180} = (A0-B0)-(A2-B2)$

$Q = c_{90} - c_{270} = (A1-B1)-(A3-B3) \quad (3)$

(49) By the 4-phase method, for example, like “A**0**-A**2**” and “A**1**-A**3**” in Formula (3), by taking the difference between the detection signals of opposite phases in the same pixel, it is possible to eliminate characteristic variation between taps existing in each pixel, that is, fixed pattern noise.

(50) Meanwhile, by the 2-phase method, the depth value d to the object can be obtained by using only two phases orthogonal to each other among the detection signals A**0** to A**3** and the detection signals B**0** to B**3** obtained by setting the phase at 0°, 90°, 180°, and 270°. For example, in a case where the detection signals A**0** and B**0** in phase 0° and the detection signals A**1** and B**1** in phase 90° are used, I and Q of Formula (2) are the following Formula (4).

$I = c_{0} - c_{180} = (A0-B0)$

$Q = c_{90} - c_{270} = (A1-B1) \quad (4)$

(51) For example, in a case where the detection signals A**2** and B**2** in phase 180° and the detection signals A**3** and B**3** in phase 270° are used, I and Q of Formula (2) are the following Formula (5).

$I = c_{0} - c_{180} = -(A2-B2)$

$Q = c_{90} - c_{270} = -(A3-B3) \quad (5)$

(52) The 2-phase method cannot eliminate the characteristic variation between taps existing in each pixel, but the depth value d to the object can be obtained only from the detection signals of two phases, and therefore it is possible to measure the distance at a frame rate twice that of the 4-phase method.

(53) The signal processing unit **15** of the distance-measuring module **11** performs signal processing for appropriately selecting or blending, according to movement of the object or the like, the I signal and the Q signal corresponding to the delay time ΔT calculated by the 4-phase method, and the I signal and the Q signal corresponding to the delay time ΔT calculated by the 2-phase method, calculates the depth value d by using the result, and outputs the depth map.

(54) <3. Output Timing of Depth Map>

(55) Next, output timing of the depth map generated by the distance-measuring module **11** will be described.

(56) FIG. **6** is a diagram showing driving of the light-receiving unit **14** of the distance-measuring module **11** and the output timing of the depth map.

(57) As described above, the light-receiving unit **14** of the distance-measuring module **11** is driven to receive the reflected light by changing the phase in the order of phase 0°, phase 90°, phase 180°, and phase 270° in a time division manner, and is driven continuously with two phases for calculating the depth value d by the 2-phase method as one set.

(58) That is, as shown in FIG. **6**, the light-receiving unit **14** continuously receives light in phase 0° and light in phase 90° from time t**1**. After the standby period from time t**2** to time t**3**, the light-

receiving unit **14** continuously receives light in phase 180° and light in phase 270° from time t**3**. Next, after the predetermined standby period from time t**4** to time t**5**, the light-receiving unit **14** continuously receives light in phase 180° and light in phase 270° from time t**5**. After the standby period from time t**6** to time t**7**, the light-receiving unit **14** continuously receives light in phase 0° and light in phase 90° from time t**7**.
(59) The light receiving operation in each phase includes a reset operation of turning on the reset transistors **44**A and **44**B to reset the charge, an integration operation of accumulating the charge in the FD units **42**A and **42**B, and a read-out operation of reading the charge accumulated in the FD units **42**A and **42**B.
(60) The signal processing unit **15** calculates the depth value by using the pixel data for four phases, and outputs the depth map in units of two phases.
(61) Specifically, the signal processing unit **15** generates and outputs the depth map Depth #**1** at time t**4** by using the pixel data for four phases from time t**1** to time t**4**. At the next time t**6**, the signal processing unit **15** generates and outputs the depth map Depth #**2** by using the pixel data for four phases from time t**3** to time t**6**. At the next time t**8**, the signal processing unit **15** generates and outputs the depth map Depth #**3** by using the pixel data for four phases from time t**5** to time t**8**.
(62) By continuous driving with the two phases as one set in this way, in a case where the object is moving, an influence of movement of the object can be suppressed in the depth value calculated by the 2-phase method.
(63) <4. Detailed Configuration Example of Signal Processing Unit>
(64) FIG. **7** is a block diagram showing a detailed configuration of the signal processing unit **15**.
(65) The signal processing unit **15** includes a correction processing unit **61**, a 2-phase processing unit **62**, a 4-phase processing unit **63**, a movement estimation unit **64**, an amplitude estimation unit **65**, a fixed pattern estimation unit **66**, a blend processing unit **67**, and a phase calculation unit **68**.
(66) The signal processing unit **15** is sequentially supplied with the detection signals A**0** to A**3** and the detection signals B**0** to B**3** of each pixel of the pixel array unit **22** from the light-receiving unit **14**. The detection signals A**0** to A**3** are the detection signals A obtained by sequentially setting the phase at 0°, 90°, 180°, and 270° in the first tap **32**A. The detection signals B**0** to B**3** are the detection signals B obtained by sequentially setting the phase at 0°, 90°, 180°, and 270° in the second tap **32**B.
(67) As described with reference to FIG. **6**, the signal processing unit **15** generates and outputs the depth map for each pixel by using the latest detection signals A**0** to A**3** and detection signals B**0** to B**3** in phase 0°, phase 90°, phase 180°, and phase 270°. A combination of the detection signals A**0** to A**3** and the detection signals B**0** to B**3** includes a case where the detection signal in two phases of phase 180° and phase 270° is the latest detection signal as shown in A of FIG. **8**, and a case where the detection signal in two phases of phase 0° and phase 90° is the latest detection signal as shown in B of FIG. **8**.
(68) In the following description, for simple description, each process of the signal processing unit **15** will be described by taking, for example, a case where the combination of the detection signals A**0** to A**3** and the detection signals B**0** to B**3** supplied from the light-receiving unit **14** to the signal processing unit **15** is that the detection signals in two phases of phase 180° and phase 270° shown in A of FIG. **8** are the latest detection signals.
(69) Note that in a case where the detection signal in two phases of phase 0° and phase 90° is the latest detection signal, as shown in B of FIG. **8**, similar processing can be performed by regarding the detection signal in phase 180°, phase 270°, phase 0°, and phase 90° as the detection signals A**0** to A**3** and the detection signals B**0** to B**3**, and inverting the signs.
(70) Returning to the description of FIG. **7**, the signal processing unit **15** sequentially performs similar processing, for each pixel as the pixel to be processed, on the detection signals A**0** to A**3** and the detection signals B**0** to B**3** of each pixel of the pixel array unit **22** supplied from the light-receiving unit **14**. Therefore, in the following, each process of the signal processing unit **15** will be

described as the process of one pixel that is the pixel to be processed.
(71) The detection signals A**0** to A**3** and the detection signals B**0** to B**3** of the predetermined pixel **21** as the pixel to be processed supplied from the light-receiving unit **14** to the signal processing unit **15** are supplied to the 4-phase processing unit **63**, the movement estimation unit **64**, the amplitude estimation unit **65**, and the fixed pattern estimation unit **66**. Furthermore, the latest detection signals A**2**, A**3**, B**2**, and B**3** in two phases of phase 180° and phase 270° are supplied to the correction processing unit **61**.
(72) The correction processing unit **61** performs processing for correcting the characteristic variation (sensitivity difference) between taps of the detection signal A of the first tap **32**A of the pixel to be processed and the detection signal B of the second tap **32**B by using correction parameters supplied from the fixed pattern estimation unit **66**.
(73) In the present embodiment, the detection signal B of the second tap **32**B of the pixel to be processed is matched with the detection signal A of the first tap **32**A, and the correction processing unit **61** performs the following correction processing on each of the detection signals A (A**2**, A**3**) and B (B**2**, B**3**) in phase 180° and phase 270.
$A'=A$
$B'=c0+c1.\text{Math}.B$ (6)
(74) Here, c**0** and c**1** are correction parameters supplied from the fixed pattern estimation unit **66**, c**0** represents an offset of the detection signal B with respect to the detection signal A, and c**1** represents a gain of the detection signal B with respect to the detection signal A.
(75) The detection signals A′ and B′ in Formula (6) represent the detection signals after the correction processing. Note that in the correction processing, the detection signal A of the first tap **32**A can be matched with the detection signal B of the second tap **32**B of the pixel to be processed, or may be matched with the middle of the detection signals A and B.
(76) The correction processing unit **61** supplies the detection signals A**2**′ and B**2**′ in phase 180° and the detection signals A**3**′ and B**3**′ in phase 270° after the correction processing to the 2-phase processing unit **62**.
(77) The 2-phase processing unit **62** calculates the I signal and the Q signal of the 2-phase method by Formula (5) by using the detection signals A**2**′ and B**2**′ in phase 180° and the detection signals A**3**′ and B**3**′ in phase 270° from the correction processing unit **61**.
(78) Note that in the following, in order to distinguish from the I signal and the Q signal of the 4-phase method calculated by the 4-phase processing unit **63**, the I signal and the Q signal of the 2-phase method are described as an I2 signal and a Q2 signal.
(79) The 2-phase processing unit **62** supplies the I2 signal and the Q2 signal of the 2-phase method calculated by Formula (5) to the blend processing unit **67**.
(80) The 4-phase processing unit **63** calculates the I signal and the Q signal of the 4-phase method by Formula (3) by using the detection signals A**0** to A**3** and the detection signals B**0** to B**3** of the pixel to be processed supplied from the light-receiving unit **14**. In the following, in order to distinguish from the I2 signal and the Q2 signal of the 2-phase method, the I signal and the Q signal of the 4-phase method are described as an I4 signal and a Q4 signal.
(81) The 4-phase processing unit **63** supplies the I4 signal and the Q4 signal of the 4-phase method calculated by Formula (3) to the blend processing unit **67**.
(82) The movement estimation unit **64** estimates (calculates) a movement amount diff of the object between sets of the set of phase 0° and phase 90° and the set of phase 180° and phase 270° by using the detection signals A**0** to A**3** and the detection signals B**0** to B**3** of the pixel to be processed.
(83) The movement estimation unit **64** can adopt either of the following diff0 to diff2 as the movement amount diff of the object between sets.
$\text{diff0}=|(A0+B0+A1+B1)-(A2+B2+A3+B3)|$
$\text{diff1}=|(A0+B0)-(A2+B2)|+|(A1+B1)-(A3+B3)|$ (7)
$\text{diff2}=\text{sqrt}(|(A0+B0)-(A2+B2)|.\text{sup}.2+|(A1+B1)-(A3+B3)|.\text{sup}.2)$

(84) diff0 is a formula for calculating the movement amount from the difference in the sum of the I signal and the Q signal between sets. diff1 is a formula for calculating the movement amount from the difference in the I signal between sets. diff2 is a formula for calculating the movement amount from the distance between sets on the IQ plane. Which of the movement amounts diff0 to diff2 is adopted may be fixedly determined or may be selected (switched) with a setting signal and the like.

(85) The movement estimation unit **64** supplies the estimated movement amount diff of the object to the fixed pattern estimation unit **66** and the blend processing unit **67**.

(86) The amplitude estimation unit **65** estimates (calculates) amplitude amp of the detection signal of the pixel to be processed supplied from the light-receiving unit **14**. The amplitude here represents the difference in the detection signal between two phases caused by the modulated emission light. A large amplitude amp indicates that the emission light is sufficiently reflected from the object and is incident on the pixel to be processed. A small amplitude amp represents a large noise.

(87) The amplitude estimation unit **65** can adopt either of the following amp0 to amp3 as the amplitude amp of the detection signal.

amp0=|A2−B2)−(A3−B3)|

amp1=sqrt(|(A2−B2)|.sup.2+|(A3−B3)|.sup.2)

amp2=|(A0−B0)−(A2−B2)|+|(A1−B1)−(A3−B3)|)   (8)

amp3=sqrt(|(A0−B0)−(A2−B2)|.sup.2+|(A1−B1)−(A3−B3)|.sup.2)

(88) amp0 and amp1 are formulas for calculating the amplitude by using only the latest detection signals in two phases, that is, phase 180° and phase 270°. amp2 and amp3 are formulas for calculating the amplitude by using the latest detection signals in four phases, that is, phase 0°, phase 90°, phase 180°, and phase 270°.

(89) The amplitude estimation unit **65** supplies the estimated amplitude amp of the detection signals to the fixed pattern estimation unit **66** and the blend processing unit **67**.

(90) The fixed pattern estimation unit **66** estimates (calculates) an offset c**0** and a gain c**1**, which are correction parameters for correcting the characteristic variation (sensitivity difference) between taps by using the detection signals A**0** to A**3** and the detection signals B**0** to B**3** of the pixel to be processed, the movement amount diff of the object supplied from the movement estimation unit **64**, and the amplitude amp supplied from the amplitude estimation unit **65**.

(91) The light receiving periods of the first tap **32**A and the second tap **32**B of the predetermined pixel **21** as the pixel to be processed are 180° out of phase. Therefore, the following relationship is established between the offset c**0** and the gain c**1** and the detection signals A**0** to A**3** and the detection signals B**0** to B**3** under ideal conditions.

B0=c0+c1.Math.A2

B1=c0+c1.Math.A3

B2=c0+c1.Math.A0   (9)

B3=c0+c1.Math.A1

(92) When the matrices A, x, and y are placed as follows,

(93) $A = \begin{bmatrix} 1 & A2 \\ 1 & A3 \\ 1 & A0 \\ 1 & A1 \end{bmatrix} x = \begin{bmatrix} c0 \\ c1 \end{bmatrix} y = \begin{bmatrix} B0 \\ B1 \\ B2 \\ B3 \end{bmatrix}$ [Formula 3]

(94) Formula (9) can be expressed as y=Ax, and thus, the matrix x, that is, the offset c**0** and the gain c**1** can be calculated by the following Formula (10) by the least squares method.

[Formula 4]

X=(A.sup.TA).sup.−1A.sup.Ty (10)

(95) More strictly, the fixed pattern estimation unit **66** retains the current offset c**0** and the gain c**1**, or updates to the newly calculated offset c**0** and the gain c**1** according to the movement amount diff

of the object supplied from the movement estimation unit **64** and the amplitude amp supplied from the amplitude estimation unit **65**. Details will be described later with reference to FIG. **10**.
(96) According to the movement amount diff and the amplitude amp, the blend processing unit **67** blends the I2 signal and the Q2 signal of the 2-phase method supplied from the 2-phase processing unit **62**, and the I4 signal and the Q4 signal of the 4-phase method supplied from the 4-phase processing unit **63**, and calculates the I signal and the Q signal after blending and supplies the I signal and the Q signal to the phase calculation unit **68**.
(97) Specifically, on the basis of the movement amount diff of the object supplied from the movement estimation unit **64**, the blend processing unit **67** calculates a blend rate α_diff based on the movement amount diff by the following Formula (11).

[Formula 5]

(98)
$$\alpha\_diff = \begin{cases} 0 & (diff < dth\,0) \\ (diff - dth\,0)/(dth\,1 - dth\,0) & (dth\,0 \leqq diff < dth\,1) \\ 1 & (dth\,1 \leqq diff) \end{cases} \quad (11)$$

(99) According to Formula (11), as shown in A of FIG. **9**, in a case where the movement amount diff supplied from the movement estimation unit **64** is smaller than a first threshold value dth**0**, the blend rate α_diff=0 is set. In a case where the movement amount diff is equal to or greater than a second threshold value dth**1**, the blend rate α_diff=1 is set. In a case where the movement amount diff is equal to or greater than the first threshold value dth**0** and less than the second threshold value dth**1**, the blend rate α_diff is linearly determined in the range of 0<α_diff<1.
(100) Furthermore, on the basis of the amplitude amp supplied from the amplitude estimation unit **65**, the blend processing unit **67** calculates the blend rate α_amp based on the amplitude amp by the following Formula (12).

[Formula 6]

(101)
$$\alpha\_amp = \begin{cases} 0 & (amp < ath\,0) \\ (amp - ath\,0)/(ath\,1 - ath\,0) & (ath\,0 \leqq amp < ath\,1) \\ 1 & (ath\,1 \leqq amp) \end{cases} \quad (12)$$

(102) According to Formula (12), as shown in B of FIG. **9**, in a case where the amplitude amp supplied from the amplitude estimation unit **65** is smaller than a first threshold value ath**0**, the blend rate α_amp=0 is set. In a case where the amplitude amp is equal to or greater than a second threshold value ath**1**, the blend rate α_amp=1 is set. In a case where the amplitude amp is equal to or greater than the first threshold value ath**0** and less than the second threshold value ath**1**, the blend rate α_amp is linearly determined in the range of 0<α_amp<1.
(103) Then, the blend processing unit **67** calculates the final blend rate α from the blend rate α_diff based on the movement amount diff and the blend rate α_amp based on the amplitude amp by either of the following Formula (12A) or (12B).
α=min(α_diff,α_amp) (12A)
α=β.Math.α_diff+(1−β)*α_amp (12B)
(104) Note that β in Formula (12B) is a blend coefficient for blending the blend rate α_diff and the blend rate α_amp, and, for example, is set in advance.
(105) The blend processing unit **67** calculates the I signal and the Q signal obtained by blending the I2 signal and the Q2 signal of the 2-phase method and the I4 signal and the Q4 signal of the 4-phase method by Formula (13) by using the calculated final blend rate α, and supplies the signals to the phase calculation unit **68**.
I=*α.Math.I*2+(1−α).Math.*I*4
Q=*α.Math.Q*2+(1−α).Math.*I*4 (13)
(106) In a case where the movement amount diff is large, the blend processing unit **67** performs

blending to increase the ratio of the I2 signal and the Q2 signal of the 2-phase method. In a case where the movement amount diff is small, the blend processing unit **67** performs blending to increase the ratio of the I4 signal and the Q4 signal of the 4-phase method with less noise. Furthermore, in a case where the amplitude amp is small (noisy), the blend processing unit **67** performs blending to increase the ratio of the I4 signal and the Q4 signal of the 4-phase method such that the SN ratio is improved.

(107) The phase calculation unit **68** of FIG. **7** uses the I signal and the Q signal supplied from the blend processing unit **67** to calculate the depth value d, which is distance information to the object, by the above Formulas (1) and (2). As described with reference to FIG. **6**, every time the detection signals A and B in two phases are updated, the phase calculation unit **68** calculates and outputs the depth value d (depth map) by using the latest detection signals A and B in four phases.

(108) FIG. **10** is a block diagram showing a detailed configuration example of the fixed pattern estimation unit **66**.

(109) The fixed pattern estimation unit **66** includes a coefficient calculation unit **81**, a coefficient update unit **82**, and a coefficient storage unit **83**.

(110) The detection signals A**0** to A**3** and the detection signals B**0** to B**3** of the pixel to be processed from the light-receiving unit **14** are supplied to the coefficient calculation unit **81**. The movement amount diff of the object from the movement estimation unit **64** and the amplitude amp of the detection signal from the amplitude estimation unit **65** are supplied to the coefficient update unit **82**.

(111) The coefficient calculation unit **81** calculates the matrix x, that is, the offset c**0** and the gain c**1** by the above Formula (10). The coefficient calculation unit **81** supplies the calculated offset c**0** and the gain c**1** to the coefficient update unit **82** as a new offset next_c**0** and a new gain next_c**1** as update candidates.

(112) On the basis of the movement amount diff of the object supplied from the movement estimation unit **64**, the coefficient update unit **82** calculates a blend rate u_diff based on the movement amount diff by the following Formula (14).

[Formula 7]

(113)
$$u\_diff = \begin{cases} 1 & (diff < uth\ 0) \\ (uth\ 1 - diff) / (uth\ 1 - uth\ 0) & (uth\ 0 \leqq diff < uth\ 1) \\ 0 & (uth\ 1 \leqq diff) \end{cases} \quad (14)$$

(114) According to Formula (14), as shown in A of FIG. **11**, in a case where the movement amount diff supplied from the movement estimation unit **64** is smaller than a first threshold value uth**0**, the blend rate u_diff=1 is set. In a case where the movement amount diff is equal to or greater than a second threshold value uth**1**, the blend rate u_diff=0 is set. In a case where the movement amount diff is equal to or greater than the first threshold value uth**0** and less than the second threshold value uth**1**, the blend rate u_diff is linearly determined in the range of 0<u_diff<1.

(115) Furthermore, on the basis of the amplitude amp supplied from the amplitude estimation unit **65**, the coefficient update unit **82** calculates the blend rate u_amp based on the amplitude amp by the following Formula (15).

[Formula 8]

(116)
$$u\_amp = \begin{cases} 0 & (amp < vth\ 0) \\ (amp - vth\ 0) / (vth\ 1 - vth\ 0) & (vth\ 0 \leqq amp < vth\ 1) \\ 1 & (vth\ 1 \leqq amp) \end{cases} \quad (15)$$

(117) According to Formula (15), as shown in B of FIG. **11**, in a case where the amplitude amp supplied from the amplitude estimation unit **65** is smaller than a first threshold value vth**0**, the blend rate u_amp=0 is set. In a case where the amplitude amp is equal to or greater than a second

threshold value vth1, the blend rate u_amp=1 is set. In a case where the amplitude amp is equal to or greater than the first threshold value vth0 and less than the second threshold value vth1, the blend rate u_am is linearly determined in the range of 0<u_amp<1.

(118) Then, the coefficient update unit 82 calculates the final blend rate u by the following Formula (16) from the blend rate u_diff based on the movement amount diff and the blend rate u_amp based on the amplitude amp.

$$u=\min(u\_\mathrm{diff},u\_\mathrm{amp}) \qquad (16)$$

(119) Using the calculated final blend rate u, the coefficient update unit 82 blends the new offset next_c0 and the new gain next_c1 from the coefficient calculation unit 81 with a current offset prev_c0 and a gain prev_c1 from the coefficient storage unit 83, and calculates the updated offset c0 and the gain c1 by the following Formula (17).

$$c0=u.\mathrm{Math}.\mathrm{next}\_c0+(1-u).\mathrm{Math}.\mathrm{prev}\_c0$$

$$c1=u.\mathrm{Math}.\mathrm{next}\_c1+(1-u).\mathrm{Math}.\mathrm{prev}\_c1 \qquad (17)$$

(120) The coefficient update unit 82 supplies the calculated updated offset c0 and the gain c1 to the correction processing unit 61 (FIG. 7) and stores the offset c0 and the gain c1 in the coefficient storage unit 83.

(121) The coefficient storage unit 83 stores the offset c0 and the gain c1 supplied from the coefficient update unit 82. Then, the offset c0 and the gain c1 stored in the coefficient storage unit 83 are supplied to the coefficient update unit 82 as the current offset prev_c0 and the gain prev_c1 before the update at the timing when the next new offset next_c0 and the new gain next_c1 are supplied from the coefficient calculation unit 81 to the coefficient update unit 82.

(122) The new offset next_c0 and the new gain next_c1 to be calculated by the coefficient calculation unit 81 can be calculated with the highest accuracy in a case where the amplitude amp is large enough and the movement amount diff is small. In a case where the amplitude amp is large enough and the movement amount diff is small, the coefficient update unit 82 performs update by increasing the blend rate of the new offset next_c0 and the new gain next_c1. In a case where the amplitude amp is small or the movement amount diff is large, the coefficient update unit 82 increases the blend rate of the current offset prev_c0 and the gain prev_c1, sets the blend rate u to retain the current offset prev_c0 and the gain prev_c1, and calculates the updated offset c0 and the gain c1.

(123) Note that in the above example, the coefficient update unit 82 updates the offset c0 and the gain c1 by using both the movement amount diff supplied from the movement estimation unit 64 and the amplitude amp supplied from the amplitude estimation unit 65, but may use only one of the movement amount diff or the amplitude amp to update the offset c0 and the gain c1. In that case, as the blend rate u of Formula (17), the blend rate u_diff based on the movement amount diff or the blend rate u_amp based on the amplitude amp is substituted.

(124) <5. Depth Value Calculation Processing of Signal Processing Unit>

(125) Next, with reference to the flowchart of FIG. 12, depth value calculation processing for calculating the depth value of the pixel to be processed by the signal processing unit 15 will be described. This processing is started, for example, when the detection signals A0 to A3 and the detection signals B0 to B3 of the predetermined pixel 21 in the pixel array unit 22 as the pixel to be processed are supplied.

(126) To begin with, in step S1, the 4-phase processing unit 63 calculates the I4 signal and the Q4 signal of the 4-phase method by Formula (3) by using the detection signals A0 to A3 and the detection signals B0 to B3 of the pixel to be processed supplied from the light-receiving unit 14. The calculated I4 signal and the Q4 signal of the 4-phase method are supplied to the blend processing unit 67.

(127) In step S2, the movement estimation unit 64 estimates the movement amount diff of the object between sets of the set of phase 0° and phase 90° and the set of phase 180° and phase 270° by using the detection signals A0 to A3 and the detection signals B0 to B3 of the pixel to be

processed. For example, one of diff0 to diff2 of Formula (7) is calculated as the movement amount diff. The movement estimation unit **64** supplies the estimated movement amount diff of the object to the fixed pattern estimation unit **66** and the blend processing unit **67**.
(128) In step S**3**, the amplitude estimation unit **65** estimates the amplitude amp of the detection signal of the pixel to be processed by calculating either of amp0 to amp3 of Formula (8). The amplitude estimation unit **65** supplies the estimated amplitude amp of the detection signals to the fixed pattern estimation unit **66** and the blend processing unit **67**.
(129) Steps S**1** to S**3** may be processed in different order or may be processed simultaneously.
(130) In step S**4**, the fixed pattern estimation unit **66** estimates the offset c**0** and the gain c**1**, which are correction parameters for correcting the characteristic variation between taps by using the detection signals A**0** to A**3** and the detection signals B**0** to B**3** of the pixel to be processed, the movement amount diff of the object supplied from the movement estimation unit **64**, and the amplitude amp of the detection signal supplied from the amplitude estimation unit **65**. The estimated offset c**0** and the gain c**1** are supplied to the correction processing unit **61**.
(131) In step S**5**, the correction processing unit **61** performs processing for correcting the characteristic variation between taps of the detection signal A of the first tap **32**A and the detection signal B of the second tap **32**B of the pixel to be processed by using the correction parameters supplied from the fixed pattern estimation unit **66**. Specifically, the correction processing unit **61** performs processing for matching the detection signal B of the second tap **32**B of the pixel to be processed with the detection signal A of the first tap **32**A by Formula (6) using the offset c**0** and the gain c**1**, which are the correction parameters supplied from the fixed pattern estimation unit **66**. The detection signals A**2**′ and B**2**′ in phase 180° after the correction processing and the detection signals A**3**′ and B**3**′ in phase 270° are supplied to the 2-phase processing unit **62**.
(132) In step S**6**, the 2-phase processing unit **62** calculates the I2 signal and the Q2 signal of the 2-phase method by Formula (5) by using the detection signals A**2**′ and B**2**′ in phase 180° after the correction processing and the detection signals A**3**′ and B**3**′ in phase 270°. The calculated I2 signal and the Q2 signal are supplied to the blend processing unit **67**.
(133) In step S**7**, according to the movement amount diff and the amplitude amp, the blend processing unit **67** blends the I2 signal and the Q2 signal of the 2-phase method supplied from the 2-phase processing unit **62**, and the I4 signal and the Q4 signal of the 4-phase method supplied from the 4-phase processing unit **63**, and calculates the I signal and the Q signal after blending and supplies the I signal and the Q signal to the phase calculation unit **68**.
(134) In step S**8**, the phase calculation unit **68** uses the I signal and the Q signal supplied from the blend processing unit **67** to calculate the depth value d to the object by the above Formulas (1) and (2), and outputs the depth value d to the subsequent stage.
(135) The processes of steps S**1** to S**8** described above are sequentially performed with each pixel of the pixel array unit **22** supplied from the light-receiving unit **14** as the pixel to be processed.
(136) Since the movement amount diff of the object and the amplitude amp of the detection signal are different for each pixel of the pixel array unit **22**, the I2 signal and the Q2 signal of the 2-phase method and the I4 signal and the Q4 signal of the 4-phase method supplied from the 4-phase processing unit **63** are blended at a blend rate α different for each pixel, and the I signal and the Q signal are calculated.
(137) By the depth value calculation processing described above, in a case where the movement amount diff of the object is large, the depth value d is calculated by giving priority to the I2 signal and the Q2 signal of the 2-phase method, and in a case where the movement amount diff is small and the object is stationary, the depth value d is calculated by giving priority to the I4 signal and the Q4 signal of the 4-phase method. Furthermore, the characteristic variation (sensitivity difference) between taps, which is fixed pattern noise, is estimated from the detection signals in four phases, and is corrected by the correction processing unit **61**. Therefore, the I2 signal and the Q2 signal of the two-phase processing can be calculated with high accuracy. With this configuration, the SN

ratio can be improved. That is, the distance measurement accuracy can be improved.
(138) Since the signal processing unit **15** outputs the depth value (depth map) every time the detection signals of two phases are received, a high frame rate can be achieved with a high SN ratio.
(139) <6. Modifications of Driving by Distance-Measuring Module>
(140) With reference to FIGS. **13** to **15**, modifications of driving of the distance-measuring module **11** will be described. In addition to the above-described driving, the distance-measuring module **11** can selectively perform driving of the following first modification to third modification.
(141) A of FIG. **13** shows the first modification of driving of the distance-measuring module **11**.
(142) In the embodiment described above, the light emission control unit **13** supplies, for example, the light emission control signal of a single frequency such as 20 MHz to the light-emitting unit **12**, and the light-emitting unit **12** emits modulated light of a single frequency to the object.
(143) In contrast, as shown in A of FIG. **13**, the light emission control unit **13** causes the light-emitting unit **12** to emit the emission light at a plurality of frequencies, and causes the light-receiving unit **14** to receive light. In A of FIG. **13**, the frequency of the modulated light to be emitted from the light-emitting unit **12** differs between "HIGH FREQ." and "LOW FREQ." "HIGH FREQ." is a high frequency such as, for example, 100 MHz, and "LOW FREQ." is a low frequency such as, for example, 20 MHz.
(144) The light-receiving unit **14** receives, in order, first modulated light emitted at a high frequency and second modulated light emitted at a low frequency in two phases, phase 0° and phase 90°. Next, the light-receiving unit **14** receives, in order, first modulated light emitted at a high frequency and second modulated light emitted at a low frequency in two phases, phase 180° and phase 270°. The method of calculating the depth value at each frequency is similar to the embodiment described above.
(145) The distance-measuring module **11** causes the light-emitting unit **12** to emit light at a plurality of frequencies, and causes the light-receiving unit **14** to receive light. The signal processing unit **15** performs the above-described depth value calculation processing. By calculating the depth value by using a plurality of frequencies, it is possible to eliminate the phenomenon (depth aliasing) in which a plurality of distances according to the modulation frequencies is determined to be the same.
(146) B of FIG. **13** shows the second modification of driving of the distance-measuring module **11**.
(147) In the above-described embodiment, the light receiving period (exposure period) in which each pixel **21** of the light-receiving unit **14** receives the modulated light is set at a single time.
(148) In contrast, as shown in B of FIG. **13**, each pixel **21** can receive the modulated light by setting a plurality of light receiving periods (exposure periods). In B of FIG. **13**, the light receiving period differs between "HIGH SENSITIVITY" and "LOW SENSITIVITY". "HIGH SENSITIVITY" is high sensitivity with the light receiving period set at a first light receiving period, and "LOW SENSITIVITY" is low sensitivity with the light receiving period set at a second light receiving period that is shorter than the first light receiving period.
(149) The light-receiving unit **14** receives, in order, the modulated light emitted at a predetermined frequency in two phases, phase 0° and phase 90°, with high sensitivity and low sensitivity. Next, the light-receiving unit **14** receives, in order, the modulated light emitted at a predetermined frequency in two phases, phase 180° and phase 270°, with high sensitivity and low sensitivity. The method of calculating the depth value at each frequency is similar to the embodiment described above.
(150) The distance-measuring module **11** causes the light-emitting unit **12** to emit light at a predetermined frequency, and causes the light-receiving unit **14** to receive light with two levels of sensitivity, high sensitivity and low sensitivity. The signal processing unit **15** performs the above-described depth value calculation processing. Light reception with high sensitivity enables measurement of a long distance, but light reception with low sensitivity is likely to cause

saturation. By calculating the depth value by using a plurality of levels of sensitivity, the distance measurement range can be expanded.
(151) In the example of B of FIG. **13**, high-sensitivity detection and low-sensitivity detection are performed in the same two phases, but high-sensitivity detection and low-sensitivity detection may be performed in two different phases. Specifically, in the first place, driving may be performed to receive light with high sensitivity in two phases, phase 0° and phase 90°, to receive light with low sensitivity in two phases, phase 180° and phase 270°, to receive light with high sensitivity in two phases, phase 180° and phase 270°, and then to receive light with low sensitivity in two phases, phase 0° and phase 90°.
(152) In both the first modification of driving described in A of FIG. **13** and the second modification of driving described in B of FIG. **13**, detection is performed in four phases of phase 0°, phase 90°, phase 180°, and phase 270° at a plurality of frequencies or with a plurality of levels of sensitivity, but only detection in two phases may be performed either at a plurality of frequencies or with a plurality of levels of sensitivity.
(153) For example, A of FIG. **14** shows an example in which, in driving at a plurality of frequencies shown in A of FIG. **13**, for the second modulated light emitted at a low frequency, light reception in two phases of phase 180° and phase 270° is omitted, and detection in four phases is performed only at a high frequency.
(154) Furthermore, B of FIG. **14** shows an example in which, in driving with a plurality of levels of sensitivity shown in B of FIG. **13**, light reception in two phases of phase 180° and phase 270° with low sensitivity is omitted, and detection in four phases is performed only with high sensitivity.
(155) In this way, the frame rate can be improved by performing detection only in two phases either at a plurality of frequencies or with a plurality of levels of sensitivity.
(156) FIG. **15** shows the third modification of driving of the distance-measuring module **11**.
(157) In the embodiment described above, all the pixels **21** of the pixel array unit **22** of the light-receiving unit **14** are driven to perform detection at predetermined timing in the same phase of either phase 0°, phase 90°, phase 180°, or phase 270°.
(158) In contrast, as shown in A of FIG. **15**, the pixels **21** of the pixel array unit **22** may be classified checkerwise into pixels **21**X and pixels **21**Y, and may be driven to perform detection in different phases between the pixels **21**X and the pixels **21**Y.
(159) For example, as shown in C of FIG. **15**, the light-receiving unit **14** of the distance-measuring module **11** can be driven such that in a certain frame period, the pixel **21**X of the pixel array unit **22** performs detection in phase 0°, while the pixel **21**Y performs detection in phase 90°, and in the next frame period, the pixel **21**X of the pixel array unit **22** performs detection in phase 180°, while the pixel **21**Y performs detection in phase 270°. Then, the depth value is calculated by the above-described depth value calculation processing by using the detection signals in four phases obtained in the two frame periods.
(160) The distance-measuring module **11** of FIG. **1** can be applied to, for example, a vehicle-mounted system mounted on a vehicle and measuring a distance to an object outside the vehicle. Furthermore, for example, the distance-measuring module **11** of FIG. **1** can be applied to a gesture recognition system and the like that measures a distance to an object including a user's hand and the like and recognizes the user's gesture on the basis of a measurement result thereof.
(161) <7. Configuration Example of Electronic Device>
(162) The distance-measuring module **11** described above can be mounted on an electronic device, for example, smartphones, tablet terminals, mobile phones, personal computers, game consoles, television receivers, wearable terminals, digital still cameras, digital video cameras, and the like.
(163) FIG. **16** is a block diagram showing a configuration example of a smartphone as an electronic device equipped with the distance-measuring module.
(164) As shown in FIG. **16**, the smartphone **101** includes a distance-measuring module **102**, an image capturing device **103**, a display **104**, a speaker **105**, a microphone **106**, a communication

module **107**, a sensor unit **108**, a touch panel **109**, and a control unit **110** connected via a bus **111**. Furthermore, the control unit **110** has functions as an application processing unit **121** and an operation system processing unit **122** by a CPU executing a program.
(165) The distance-measuring module **11** of FIG. **1** is applied to the distance-measuring module **102**. For example, the distance-measuring module **102** is disposed on the front of the smartphone **101** and performs distance measurement on the user of the smartphone **101**, thereby outputting the depth value of the surface shape of the user's face, hand, finger, and the like as a distance measurement result.
(166) The image capturing device **103** is disposed on the front of the smartphone **101** and captures an image of the user of the smartphone **101** as a subject, thereby acquiring an image in which the user is captured. Note that although not illustrated, a configuration in which the image capturing device **103** is also disposed on the back of the smartphone **101** may be adopted.
(167) The display **104** displays an operation screen for performing processing by the application processing unit **121** and the operation system processing unit **122**, an image captured by the image capturing device **103**, and the like. The speaker **105** and the microphone **106**, for example, output the voice of the other party and pick up the voice of the user when talking over the smartphone **101**.
(168) The communication module **107** performs communication via a communication network. The sensor unit **108** senses speed, acceleration, proximity, and the like. The touch panel **109** acquires a touch operation by the user on the operation screen displayed on the display **104**.
(169) The application processing unit **121** performs processing for providing various services by the smartphone **101**. For example, the application processing unit **121** can perform processing for creating a face that virtually reproduces the facial expression of the user by computer graphics on the basis of the depth supplied from the distance-measuring module **102**, and displaying the face on the display **104**. Furthermore, the application processing unit **121** can perform, for example, processing for creating three-dimensional shape data of an arbitrary three-dimensional object on the basis of the depth supplied from the distance-measuring module **102**.
(170) The operation system processing unit **122** performs processing for implementing basic functions and operations of the smartphone **101**. For example, the operation system processing unit **122** can perform processing for authenticating the user's face and unlocking the smartphone **101** on the basis of the depth value supplied from the distance-measuring module **102**. Furthermore, the operation system processing unit **122** can perform, for example, processing for recognizing the gesture of the user, and inputting various operations according to the gesture on the basis of the depth value supplied from the distance-measuring module **102**.
(171) The smartphone **101** configured in this way can achieve, for example, an improvement in a frame rate, a reduction in power consumption, and a reduction in a data transfer band by applying the distance-measuring module **11** described above. With this configuration, the smartphone **101** can create a face that moves more smoothly by computer graphics, perform facial authentication with high accuracy, reduce battery consumption, and transfer data in a narrow band.
(172) <8. Configuration Example of Computer>
(173) Next, a series of processes described above can be performed by hardware, or can be performed by software. In a case where the series of processes is performed by software, a program constituting the software is installed in a general purpose computer and the like.
(174) FIG. **17** is a block diagram showing a configuration example of one embodiment of the computer in which the program that performs the above-described series of processes is installed.
(175) In the computer, a central processing unit (CPU) **201**, a read only memory (ROM) **202**, a random access memory (RAM) **203**, and an electronically erasable and programmable read only memory (EEPROM) **204** are interconnected by a bus **205**. An input-output interface **206** is further connected to the bus **205**, and the input-output interface **206** is connected to the outside.
(176) In the computer configured as described above, the CPU **201** loads, for example, a program stored in the ROM **202** and the EEPROM **204** into the RAM **203** via the bus **205** and executes the

program, whereby the above-described series of processes is performed. Furthermore, the program to be executed by the computer (CPU **201**) can be installed or updated in the EEPROM **204** from the outside via the input-output interface **206** in addition to being written in the ROM **202** in advance.
(177) With this configuration, the CPU **201** performs processing according to the flowchart described above, or processing to be performed according to the configuration of the block diagram described above. Then, the CPU **201** can output processing results thereof to the outside via, for example, the input-output interface **206** as necessary.
(178) In the present specification, the processing performed by the computer according to the program does not necessarily have to be performed in chronological order in the order described as the flowchart. That is, the processing performed by the computer according to the program also includes processing performed in parallel or individually (for example, parallel processing or processing by an object).
(179) Furthermore, the program may be processed by one computer (processor) or may be processed in a distributed manner by a plurality of computers. Moreover, the program may be transferred to a distant computer and executed.
(180) <9. Example of Application to Mobile Object>
(181) The technology according to the present disclosure (present technology) can be applied to various products. For example, the technology according to the present disclosure may be implemented as a device mounted on any type of mobile object including automobiles, electric vehicles, hybrid electric vehicles, motorcycles, bicycles, personal mobility, airplanes, drones, ships, robots, and the like.
(182) FIG. **18** is a block diagram showing a schematic configuration example of a vehicle control system, which is one example of a mobile object control system to which the technology according to the present disclosure can be applied.
(183) The vehicle control system **12000** includes a plurality of electronic control units connected via a communication network **12001**. In the example shown in FIG. **18**, the vehicle control system **12000** includes a drive system control unit **12010**, a body system control unit **12020**, an outside vehicle information detection unit **12030**, an inside vehicle information detection unit **12040**, and an integrated control unit **12050**. Furthermore, as a functional configuration of the integrated control unit **12050**, a microcomputer **12051**, a voice and image output unit **12052**, and a vehicle-mounted network interface (I/F) **12053** are illustrated.
(184) The drive system control unit **12010** controls operations of devices related to the vehicle drive system according to various programs. For example, the drive system control unit **12010** functions as a control device of a driving force generation device for generating driving force of a vehicle such as an internal combustion engine or a driving motor, a driving force transmission mechanism for transmitting driving force to wheels, a steering mechanism for adjusting a steering angle of a vehicle, a braking device for generating braking force of a vehicle, and the like.
(185) The body system control unit **12020** controls operations of various devices mounted on a vehicle body according to various programs. For example, the body system control unit **12020** functions as a control device of a keyless entry system, a smart key system, a power window device, or various lamps including a head lamp, a back lamp, a brake lamp, a direction indicator, a fog lamp, and the like. In this case, radio waves transmitted from a portable device replacing a key or signals from various switches can be input into the body system control unit **12020**. The body system control unit **12020** receives the input of these radio waves or signals and controls a door lock device, the power window device, the lamps, and the like of the vehicle.
(186) The outside vehicle information detection unit **12030** detects information outside the vehicle on which the vehicle control system **12000** is mounted. For example, an image capturing unit **12031** is connected to the outside vehicle information detection unit **12030**. The outside vehicle information detection unit **12030** causes the image capturing unit **12031** to capture an image

outside the vehicle, and receives the captured image. The outside vehicle information detection unit **12030** may perform object detection processing or distance detection processing on a person, a car, an obstacle, a sign, a character on a road surface, and the like on the basis of the received image.
(187) The image capturing unit **12031** is an optical sensor that receives light and outputs an electric signal according to an amount of the received light. The image capturing unit **12031** can output the electric signal as an image or output the electric signal as distance measurement information. Furthermore, the light received by the image capturing unit **12031** may be visible light or invisible light including infrared rays and the like.
(188) The inside vehicle information detection unit **12040** detects information inside the vehicle. For example, a driver status detection unit **12041** that detects the status of a driver is connected to the inside vehicle information detection unit **12040**. The driver status detection unit **12041** may include, for example, a camera that captures images of the driver. The inside vehicle information detection unit **12040** may calculate the degree of fatigue or concentration of the driver on the basis of the detection information input from the driver status detection unit **12041**, or determine that the driver is not dozing.
(189) On the basis of information inside and outside the vehicle acquired by the outside vehicle information detection unit **12030** or the inside vehicle information detection unit **12040**, the microcomputer **12051** can calculate a control target value for the driving force generating device, the steering mechanism, or the braking device, and output a control command to the drive system control unit **12010**. For example, the microcomputer **12051** can perform cooperative control aimed at implementing functions of an advanced driver assistance system (ADAS) including vehicle collision avoidance or impact mitigation, follow-up driving based on distance between vehicles, driving while maintaining vehicle speed, vehicle collision warning, vehicle lane deviation warning, or the like.
(190) Furthermore, by controlling the driving force generating device, the steering mechanism, the braking device, or the like on the basis of information around the vehicle acquired by the outside vehicle information detection unit **12030** or the inside vehicle information detection unit **12040**, the microcomputer **12051** can perform cooperative control aimed at automatic driving and the like in which the vehicle travels autonomously without depending on the operation of the driver.
(191) Furthermore, the microcomputer **12051** can output a control command to the body system control unit **12020** on the basis of information outside the vehicle acquired by the outside vehicle information detection unit **12030**. For example, the microcomputer **12051** can perform cooperative control aimed at preventing glare such as controlling a headlamp according to the position of a preceding vehicle or an oncoming vehicle detected by the outside vehicle information detection unit **12030** and switching a high beam to a low beam.
(192) The voice and image output unit **12052** transmits an output signal of at least one of a voice or an image to an output device that can visually or auditorily notifying an occupant of the vehicle or the outside of the vehicle of information. In the example of FIG. **18**, an audio speaker **12061**, a display unit **12062**, and an instrument panel **12063** are illustrated as the output device. The display unit **12062** may include, for example, at least one of an on-board display and a head-up display.
(193) FIG. **19** is a diagram showing an example of an installation position of the image capturing unit **12031**.
(194) In FIG. **19**, as the image capturing unit **12031**, the vehicle **12100** includes image capturing units **12101**, **12102**, **12103**, **12104**, and **12105**.
(195) The image capturing units **12101**, **12102**, **12103**, **12104**, and **12105** are provided, for example, at positions such as a front nose, side mirrors, a rear bumper, a back door, and an upper part of a windshield in the vehicle compartment of the vehicle **12100**. The image capturing unit **12101** provided in the front nose and the image capturing unit **12105** provided in the upper part of the windshield in the vehicle compartment mainly acquire an image ahead of the vehicle **12100**. The image capturing units **12102** and **12103** provided in the side mirrors mainly acquire side

images of the vehicle **12100**. The image capturing unit **12104** provided in the rear bumper or back door mainly acquires an image behind the vehicle **12100**. A front image captured by the image capturing units **12101** and **12105** is mainly used for detecting a preceding vehicle, a pedestrian, an obstacle, a traffic signal, a traffic sign, a lane, or the like.
(196) Note that FIG. **19** shows one example of image capturing ranges of the image capturing units **12101** to **12104**. The image capturing range **12111** indicates the image capturing range of the image capturing unit **12101** provided in the front nose. The image capturing ranges **12112** and **12113** indicate the image capturing ranges of the image capturing units **12102** and **12103** provided in the side mirrors, respectively. The image capturing range **12114** indicates the image capturing range of the image capturing unit **12104** provided in the rear bumper or the back door. For example, image data captured by the image capturing units **12101** to **12104** is superimposed, whereby a bird's-eye view image of the vehicle **12100** viewed from above is obtained.
(197) At least one of the image capturing units **12101** to **12104** may have a function of acquiring distance information. For example, at least one of the image capturing units **12101** to **12104** may be a stereo camera including a plurality of image capturing elements or an image capturing element having pixels for detecting a phase difference.
(198) For example, on the basis of the distance information obtained from the image capturing units **12101** to **12104**, the microcomputer **12051** determines the distance to each three-dimensional object in the image capturing ranges **12111** to **12114** and the temporal change of the distance (relative speed with respect to the vehicle **12100**), thereby particularly extracting, as a preceding vehicle, a three-dimensional object that is closest on a traveling path of the vehicle **12100** and travels at a predetermined speed (for example, 0 km/h or more) in the substantially same direction as the vehicle **12100**. Moreover, the microcomputer **12051** can set a distance between vehicles to be secured before a preceding vehicle, and perform automatic brake control (including follow-up stop control), automatic acceleration control (including follow-up start control), and the like. In this way, cooperative control aimed at automatic driving and the like in which the vehicle autonomously travels without depending on the operation of the driver can be performed.
(199) For example, on the basis of the distance information obtained from the image capturing units **12101** to **12104**, the microcomputer **12051** classifies and extracts three-dimensional object data regarding three-dimensional objects into two-wheeled vehicles, ordinary vehicles, large vehicles, pedestrians, utility poles, and other three-dimensional objects, and can use the data for automatic obstacle avoidance. For example, the microcomputer **12051** discriminates obstacles around the vehicle **12100** between obstacles that can be visually recognized by the driver of the vehicle **12100** and obstacles that are difficult to be visually recognized. Then, the microcomputer **12051** determines the collision risk indicating the risk of collision with each obstacle. When the collision risk is equal to or higher than a set value and there is a possibility of collision, the microcomputer **12051** can provide driving assistance for collision avoidance by outputting an alarm to the driver via the audio speaker **12061** or the display unit **12062**, or by performing forced deceleration or avoidance steering via the drive system control unit **12010**.
(200) At least one of the image capturing units **12101** to **12104** may be an infrared camera that detects infrared rays. For example, the microcomputer **12051** can recognize a pedestrian by determining whether or not a pedestrian exists in captured images of the image capturing units **12101** to **12104**. Such pedestrian recognition is performed, for example, by a procedure for extracting feature points in the captured images of the image capturing units **12101** to **12104** serving as infrared cameras, and a procedure for performing pattern matching processing on a series of feature points indicating an outline of an object to determine whether or not the object is a pedestrian. When the microcomputer **12051** recognizes a pedestrian by determining that the pedestrian exists in the captured images of the image capturing units **12101** to **12104**, the voice and image output unit **12052** controls the display unit **12062** such that a rectangular outline for emphasis is superimposed on the recognized pedestrian. Furthermore, the voice and image output

unit **12052** may control the display unit **12062** to display an icon and the like indicating the pedestrian at a desired position.

(201) One example of the vehicle control system to which the technology according to the present disclosure can be applied has been described above. The technology according to the present disclosure can be applied to the outside vehicle information detection unit **12030** and the inside vehicle information detection unit **12040** among the configurations described above. Specifically, by using the distance measurement by the distance-measuring module **11** as the outside vehicle information detection unit **12030** or the inside vehicle information detection unit **12040**, it is possible to perform processing for recognizing the driver's gesture, perform various operations (for example, audio system, navigation system, air conditioning system) according to the gesture, and detect the driver's state more accurately. Furthermore, by using the distance measurement by the distance-measuring module **11**, it is possible to recognize unevenness of a road surface and reflect the unevenness on control of a suspension.

(202) Note that the present technology can be applied to the method of performing amplitude modulation on light projected onto an object, which is called continuous-wave method, out of the indirect ToF method. Furthermore, as the structure of the photodiode **31** of the light-receiving unit **14**, the present technology can be applied to a distance-measuring sensor having a structure to distribute charges to two charge storage units, such as a current assisted photonic demodulator (CAPD) structure distance measuring sensor or a gate method distance-measuring sensor that alternately adds pulses and charges of the photodiode to the two gates.

(203) The embodiment of the present technology is not limited to the embodiment described above, and various modifications may be made without departing from the spirit of the present technology.

(204) The plurality of present technologies described above in this specification can be independently implemented independently as long as there is no contradiction. Of course, arbitrary plurality of the present technologies can be used in combination. For example, some or all of the present technology described in either of the embodiments can be combined with some or all of the present technology described in the other embodiment. Furthermore, it is also possible to implement some or all of the arbitrary present technologies described above in combination with other technologies not described above.

(205) Furthermore, for example, the configuration described as one device (or processing unit) may be divided and configured as a plurality of devices (or processing units). On the contrary, the configurations described above as a plurality of devices (or processing units) may be collectively configured as one device (or processing unit). Furthermore, of course, a configuration other than the configuration described above may be added to the configuration of each device (or each processing unit). Moreover, if the configuration or operation of the entire system is substantially the same, a part of the configuration of one device (or processing unit) may be included in the configuration of another device (or other processing unit).

(206) Moreover, in the present specification, the system means a set of a plurality of components (devices, modules (parts), and the like), and it does not matter whether or not all the components are in the same housing. Therefore, a plurality of devices housed in separate enclosures and connected via a network, and one device in which a plurality of modules is housed in one enclosure are a system.

(207) Furthermore, for example, the program described above can be executed in an arbitrary device. In that case, the device is required at least to have necessary functions (functional blocks and the like) to be able to obtain necessary information.

(208) It should be noted that effects described in the present specification are merely illustrative and not restrictive, and effects other than those described in the present specification may be produced.

(209) Note that the present technology can have the following configurations.

(210) (1)

(211) A signal processing device including an estimation unit that estimates, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(2)
(212) The signal processing device according to (1) described above, in which the estimation unit calculates an offset and a gain of the second tap with respect to the first tap as the sensitivity difference between taps.
(3)
(213) The signal processing device according to (2) described above, in which the estimation unit calculates the offset and the gain of the second tap with respect to the first tap on condition that phases of the first tap and the second tap are 180 degrees out of phase.
(4)
(214) The signal processing device according to (2) or (3) described above, further including an amplitude estimation unit that estimates amplitude of the first to fourth detection signals, in which the estimation unit updates the offset and the gain by blending the calculated offset and the gain with the current offset and the gain on the basis of the estimated amplitude.
(5)
(215) The signal processing device according to any one of (2) to (4) described above, further including a movement amount estimation unit that estimates a movement amount of the object in the pixel, in which the estimation unit updates the offset and the gain by blending the calculated offset and the gain with the current offset and the gain on the basis of the estimated amplitude and the movement amount.
(6)
(216) The signal processing device according to any one of (1) to (5) described above, further including a correction processing unit that performs correction processing for correcting the first and second detection signals that are latest two of the first to fourth detection signals by using a parameter with which the sensitivity difference is estimated.
(7)
(217) The signal processing device according to (6) described above, further including: a 2-phase processing unit that calculates an I signal and a Q signal of a 2-phase method by using the first and second detection signals after the correction processing; a 4-phase processing unit that calculates an I signal and a Q signal of a 4-phase method by using the first to fourth detection signals; a blend processing unit that blends the I signal and the Q signal of the 2-phase method with the I signal and the Q signal of the 4-phase method, and calculates an I signal and a Q signal after the blending; and a calculation unit that calculates distance information to the object on the basis of the I signal and the Q signal after the blending.
(8)
(218) The signal processing device according to (7) described above, in which the blend processing unit blends the I signal and the Q signal of the 2-phase method with the I signal and the Q signal of the 4-phase method on the basis of amplitude of the first to fourth detection signals and a movement amount of the object in the pixel.
(9)
(219) The signal processing device according to (7) or (8) described above, in which the calculation unit calculates the distance information to the object every time detection signals of two phases of the first to fourth detection signals are updated.
(10)
(220) A signal processing method including, by a signal processing device: estimating, in a pixel

including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using a first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(11)
(221) A distance-measuring module including a light-receiving unit in which pixels each including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit are two-dimensionally arranged; and a signal processing unit including an estimation unit that estimates a sensitivity difference between taps of the first tap and the second tap in the pixels by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light.
(12)
(222) The distance-measuring module according to (11) described above, in which each of the pixels receives the reflected light obtained by emitting the emission light at a plurality of frequencies, and the estimation unit estimates the sensitivity difference between taps at each of the plurality of frequencies.
(13)
(223) The distance-measuring module according to (11) or (12) described above, in which each of the pixels receives the reflected light obtained by emitting the emission light at a plurality of exposure times, and the estimation unit estimates the sensitivity difference between taps at each of the plurality of exposure times.
(14)
(224) The distance-measuring module according to any one of (11) to (13) described above, in which the light-receiving unit is driven to cause a first pixel to receive the reflected light in the first phase and to cause a second pixel to receive the reflected light in the second phase simultaneously, next, to cause the first pixel to receive the reflected light in the third phase and to cause the second pixel to receive the reflected light in the fourth phase simultaneously, and the estimation unit estimates the sensitivity difference between taps of the first tap and the second tap by using the first to fourth detection signals detected in the first to fourth phases.
REFERENCE SIGNS LIST
(225) **11** Distance-measuring module **13** Light emission control unit **14** Light-receiving unit **15** Signal processing unit **21** Pixel **18** Reference signal generation unit **18***a*, **18***b* DAC **18***c* Control unit **21** Pixel **22** Pixel array unit **23** Drive control circuit **31** Photodiode **32**A First tap **32**B Second tap **61** Correction processing unit **62** 2-phase processing unit **63** 4-phase processing unit **64** Movement estimation unit **65** Amplitude estimation unit **66** Fixed pattern estimation unit **67** Blend processing unit **68** Phase calculation unit **81** Coefficient calculation unit **82** Coefficient update unit **83** Coefficient storage unit **101** Smartphone **102** Distance-measuring module **201** CPU **202** ROM **203** RAM

Claims

1. A signal processing device comprising an estimation unit that estimates, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light, wherein the estimation unit calculates an offset and a gain of the second tap with respect to the first tap as the sensitivity

difference between taps, and wherein the estimation unit is implemented via at least one processor.
2. The signal processing device according to claim 1, wherein the estimation unit calculates the offset and the gain of the second tap with respect to the first tap on condition that phases of the first tap and the second tap are 180 degrees out of phase.
3. The signal processing device according to claim 1, further comprising an amplitude estimation unit that estimates amplitude of the first to fourth detection signals, wherein the estimation unit updates the offset and the gain by blending the calculated offset and the gain with the current offset and the gain on a basis of the estimated amplitude, and wherein the amplitude estimation unit is implemented via at least one processor.
4. The signal processing device according to claim 3, further comprising a movement amount estimation unit that estimates a movement amount of the object in the pixel, wherein the estimation unit updates the offset and the gain by blending the calculated offset and the gain with the current offset and the gain on a basis of the estimated amplitude and the movement amount, and wherein the movement amount estimation unit is implemented via at least one processor.
5. The signal processing device according to claim 1, further comprising a correction processing unit that performs correction processing for correcting the first and second detection signals that are latest two of the first to fourth detection signals by using a parameter with which the sensitivity difference is estimated, and wherein the correction processing unit is implemented via at least one processor.
6. A signal processing device comprising: an estimation unit that estimates, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light, a 2-phase processing unit that calculates an I signal and a Q signal of a 2-phase method by using the first and second detection signals after the correction processing; a 4-phase processing unit that calculates an I signal and a Q signal of a 4-phase method by using the first to fourth detection signals; a blend processing unit that blends the I signal and the Q signal of the 2-phase method with the I signal and the Q signal of the 4-phase method, and calculates an I signal and a Q signal after the blending; and a calculation unit that calculates distance information to the object on a basis of the I signal and the Q signal after the blending, wherein the estimation unit, the 2-phase processing unit, the 4-phase processing unit, the blend processing unit, and the calculation unit are each implemented via at least one processor.
7. The signal processing device according to claim 6, wherein the blend processing unit blends the I signal and the Q signal of the 2-phase method with the I signal and the Q signal of the 4-phase method on a basis of amplitude of the first to fourth detection signals and a movement amount of the object in the pixel.
8. The signal processing device according to claim 6, wherein the calculation unit calculates the distance information to the object every time detection signals of two phases of the first to fourth detection signals are updated.
9. A signal processing method comprising, by a signal processing device: estimating, in a pixel including a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit, a sensitivity difference between taps of the first tap and the second tap by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light; and calculating an offset and a gain of the second tap with respect to the first tap as the sensitivity difference between taps.
10. A distance-measuring module comprising a light-receiving unit in which pixels each including

a first tap that detects a charge photoelectrically converted by a photoelectric conversion unit and a second tap that detects the charge photoelectrically converted by the photoelectric conversion unit are two-dimensionally arranged; and a signal processing unit including an estimation unit that estimates a sensitivity difference between taps of the first tap and the second tap in the pixels by using first to fourth detection signals obtained by detecting reflected light generated by emission light reflected by an object in first to fourth phases with respect to the emission light, wherein the light-receiving unit is driven to cause a first pixel to receive the reflected light in the first phase and to cause a second pixel to receive the reflected light in the second phase simultaneously, next, to cause the first pixel to receive the reflected light in the third phase and to cause the second pixel to receive the reflected light in the fourth phase simultaneously, wherein the estimation unit estimates the sensitivity difference between taps of the first tap and the second tap by using the first to fourth detection signals detected in the first to fourth phases, and wherein the signal processing unit is implemented via at least one processor.

11. The distance-measuring module according to claim 10, wherein each of the pixels receives the reflected light obtained by emitting the emission light at a plurality of frequencies, and the estimation unit estimates the sensitivity difference between taps at each of the plurality of frequencies.

12. The distance-measuring module according to claim 10, wherein each of the pixels receives the reflected light obtained by emitting the emission light at a plurality of exposure times, and the estimation unit estimates the sensitivity difference between taps at each of the plurality of exposure times.